# FUNDAMENTALS OF SOFTWARE TESTING

## Software testing – Introduction:

- Software testing is the process of evaluating and verifying that a software product or application functions correctly, securely and efficiently according to its specific requirements.
- The main goal of testing is to improve quality and reduce risk before releasing the product to users.
- Testing can be performed manually or using automation tools like Selenium, Cypress, PyTest, etc.

## Types of testing:

The two main types of testing are manual testing and automation testing.

i. **Manual testing:**

- Manual testing is a technique to test the software that is carried out using the functions and features of an application by the tester.
- The tester manually executes test cases and verifies results.

**Features:**

- No scripting is required.
- Efficient for unplanned changes.
- Good for exploratory testing.

**Example:**

Manually testing a login page by entering valid and invalid credentials.

**ii. Automation testing:**

- Automation testing is a technique where the Tester writes scripts independently and uses suitable Software or Automation Tools to test the software.
- It allows for executing repetitive tasks without the use of a Manual Tester.

**Features:**

- Simplifies test case execution.
- Improves reliability of tests.
- Increases amount of test coverage.
- Minimizes human interaction.

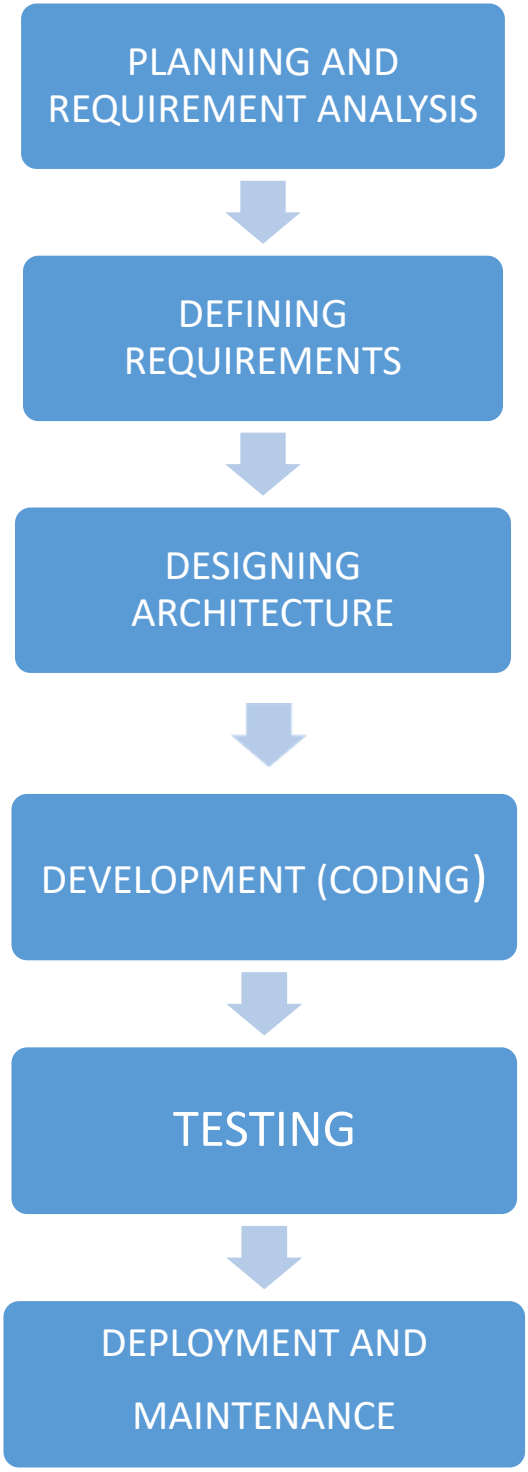
**Example:**

Using Selenium to automatically test login functionality multiple times.

**Software Development Life Cycle(SDLC):**

- The Software Development Life Cycle (SDLC) is a structured framework used by software organizations to design, develop, and test high-quality software**.**
- It defines the entire process of software production, from the initial idea to the final deployment and maintenance
- Without a structured SDLC, software development becomes chaotic. SDLC provides:
    - Clear process structure.
    - Better project plan.
    - Quality control.
    - Risk management.
    - Cost estimation.

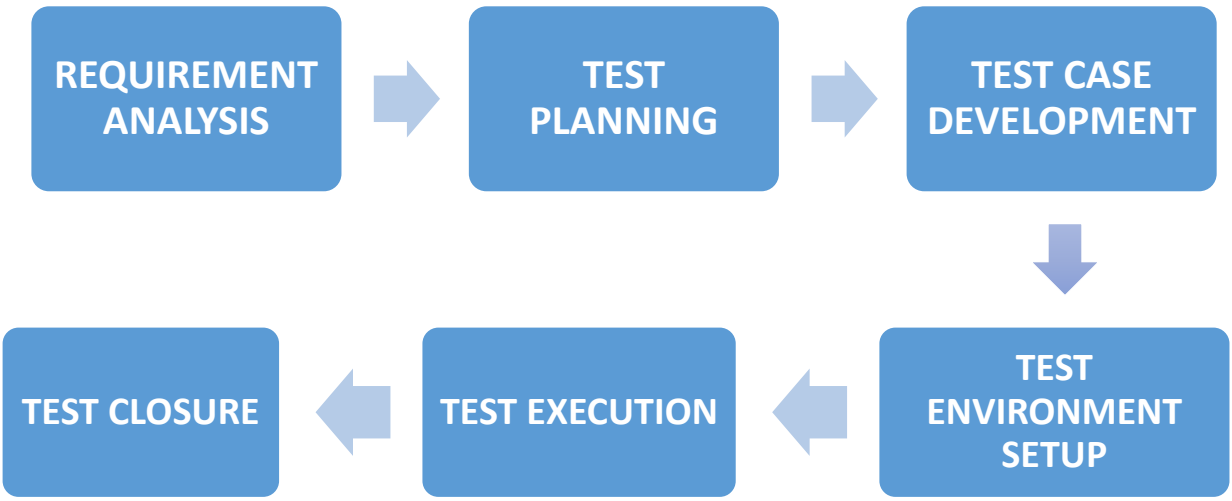
**Stages of Software Development Life Cycle (SDLC):**



**Software Testing Life Cycle (STLC):**

- The Software Testing Life Cycle (STLC) is a process that verifies whether the Software Quality meets the expectations or not.
- Software Testing Life Cycle (STLC) is a fundamental part of the Software Development Life Cycle (SDLC).

**Phases of Software Testing Life Cycle (STLC):**




**Severity in testing:**

- Severity refers to the impact of a defect on the system.
- A higher effect of the bug on system functionality will lead to a higher severity level.
- A QA engineer determines the severity level of a bug based on the technical impact.

**Severity levels:**

| Severity level | Impact | Example |
|---|---|---|
| Critical | System crashes | Payment page crashes |
| High | Major function fails | Login not working |

| | | |
|---|---|---|
| Medium | Partial issue | Slow search results |
| Low | Cosmetic issue | Typo in footer |

**Priority in testing:**

- Priority refers to how urgently a defect needs to be fixed.
- Defects having a higher priority should be fixed first.
- It is usually decided by the project manager or client based on business importance.

**Priority levels:**

| Priority level | Urgency | Example |
|---|---|---|
| High | Immediate fix required | Login not working |
| Medium | Fix soon | Profile update issue |
| Low | Can wait | Minor UI misalignment |

**Conclusion:**

Software testing ensures that software is developed in a structured manner and meets quality standards. Concepts like SDLC, STLC, severity, and priority help teams manages development and defects effectively. A strong testing process ultimately leads to reliable software and better user satisfaction.